2.1 Camera Raw

Even though it's sometimes called RAW, the camera raw image format isn't an acronym—it's literal. A raw file contains all the unprocessed data collected by your camera's sensors. It's the most accurate representation of the photograph you took, and contains more colour information than common image formats—specifically the JPEG format that most digital cameras use. Only high-end cameras record raw data (the rest convert the photo to JPEG before writing it to the memory card), so you'll probably not work with this format unless you own a Digital SLR (DSLR).

Unlike standard formats like JPEG or TIFF, however, camera raw formats are specific to manufacturers, and would originally work only with software that was provided by them. Unfortunately, this software would invariably be badly designed, leading first to open source software like dcraw that reverse-engineered the proprietary raw formats, and then the introduction of Adobe's DNG (Digital Negative) image format—a standard raw format for all cameras. Eventually, camera vendors relented, and while they didn't adopt the DNG format, they did allow third-party software to be able to read their raw formats.

For all practical purposes, a raw file is the digital equivalent of the film negative, and just like you can use chemicals to develop the film the way you want, you can use software to "develop" your camera's raw data to a format like TIFF. In either case, the negative remains intact, so you can start over if you don't like the results.

RAW Image Formats

.raf	Fuji
.crw, .cr2	Canon
.tif .kdc .dcr	Kodak
.mrw	Minolta
.nef	Nikon
.orf	Olympus
.dng	Adobe
.ptx, .pef	Pentax
.arw, .srf, .sr2	Sony
.raw	Panasonic

2.1.1 Camera Raw In Photoshop

Adobe Photoshop supports the raw formats for nearly all popular digital camera manufacturers; you'll have to download the